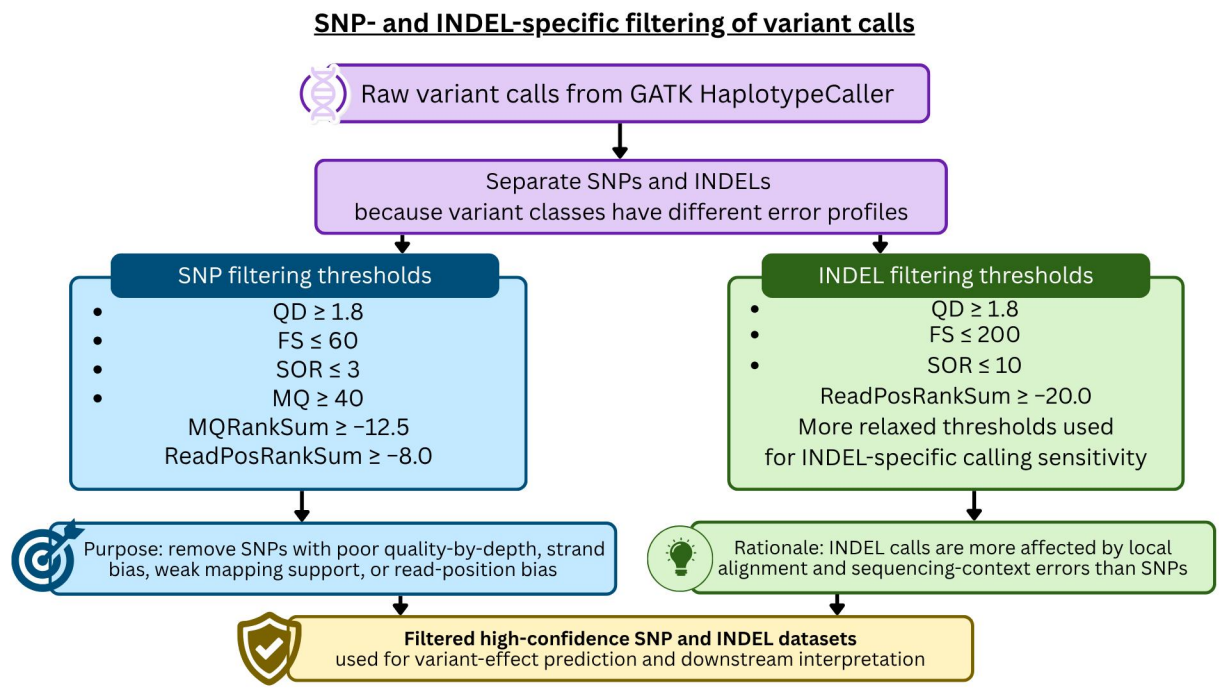

SNP- and INDEL-specific filtering of variant calls
Raw variant calls from GATK HaplotypeCaller
Separate SNPs and INDELs
because variant classes have different error profiles
SNP filtering thresholds
QD ≥ 1.8
FS ≤ 60
SOR ≤ 3
MQ ≥ 40
MQRankSum ≥ −12.5
ReadPosRankSum ≥ −8.0
INDEL filtering thresholds
QD ≥ 1.8
FS ≤ 200
SOR ≤ 10
ReadPosRankSum ≥ −20.0
More relaxed thresholds used
for INDEL-specific calling sensitivity
Purpose: remove SNPs with poor quality-by-depth, strand bias, weak mapping support, or read-position bias
Rationale: INDEL calls are more affected by local alignment and sequencing-context errors than SNPs
Filtered high-confidence SNP and INDEL datasets
used for variant-effect prediction and downstream interpretation